Towards the Prediction of Development Effort for Hypermedia Applications

Emilia Mendes
Computer Science Department
The University of Auckland
Auckland, New Zealand
emilia@cs.auckland.ac.nz

Steve Counsell
Computer Science Department
Birkbeck College
London, UK
steve@dcs.bbk.ac.uk

Nile Mosley
Computer Science Department
Auckland University of Technology,
Auckland, New Zealand
nile_mosley@yahoo.com

ABSTRACT
Accurate estimates of development effort play an important role in the successful management of larger hypermedia development projects. By applying measurement principles to measure characteristics of the applications and their development processes, feedback can be obtained to help understand, control and improve future applications and corresponding processes.

The objective of this paper is to describe the application of case-based reasoning to estimating the effort for developing hypermedia applications. The data used in the estimation process was obtained through an experiment where effort and size metrics were collected.

We have shown that case-based reasoning is a candidate technique to effort estimation and, with the aid of an automated environment, can be applied to hypermedia development effort prediction.

KEYWORDS: effort prediction, hypermedia development, case-based reasoning.

INTRODUCTION

In any scientific field, measurement generates quantitative descriptions of processes and products, making it possible to understand behaviour and result. This understanding allows for the selection of better techniques and tools to control and improve processes, products and resources [19].

Software practitioners recognise the importance of realistic estimates of effort for the successful management of large software projects, hypermedia applications are no exception.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage, and that copies bear this notice and the full citation on the first page. To copy otherwise, to republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee.
HT'01 8/01 Aarhus, Denmark
® 2001 ACM ISBN 1-59113-420-7/01/0008…$5.00

Using realistic estimates at an early stage in a project's life cycle allow project managers and development organisations to manage resources effectively.

Prediction, a necessary part of an effective process, be it authoring, design, testing, or development, comprises [5]:

step 1. Capturing data from past projects (software and development processes) which can be used to estimate effort.
step 2. Generation of prediction models based on historical data, collected during step 1.
step 3. Evaluation of how effective those models are.

This paper focuses on effort prediction for design and authoring processes. We adopt the classification proposed by Lowe and Hall [13] where authoring encompasses the management of activities for the actual content and structure of the application and its presentation. Design covers the methods used for generating the structure and functionality of the application, and typically does not include aspects such as application requirements elicitation, feasibility consideration and applications maintenance. In addition, our design phase also incorporates the conceptual design of the application, reflected in map diagrams showing documents and links.

As effort is the largest and least predictable component of project costs, most approaches found in the literature focus on this measure [5]. In this paper we only address design and authoring effort.

The data used in the estimation process was gathered through an experiment [6] where a set of suggested metrics for effort prediction were measured. The metrics were classified as size, effort and confounding factors [12].

A number of size metrics from the literature in Software Engineering and Multimedia were reused [5,9,25] or adapted [15]. The effort for authoring and designing hypermedia applications was estimated using Case-based reasoning (CBR) [11,21] also called estimation by analogy.

The rationale for CBR is to characterise the project, for which the estimate is to be made, relative to a number of attributes (e.g. application complexity, link complexity etc). This description is then used to find other similar projects, already finished, and an estimate for the new project is made based on the known effort values for those finished projects.

This paper is organised as follows: Section 2 presents related work in development effort prediction for hypermedia applications. Section 3 presents in more depth the technique of case-based reasoning and in Section 4 we describe the tool used to automate the estimation process. In Section 5 we present the experiment and Section 6 describes the empirical results of applying CBR. Finally, Section 7 concludes with an analysis of the strengths and weaknesses of using CBR and suggestions on future work.

RELATED WORK

To date there is a little work on the development effort prediction for hypermedia applications, with all results based on Web applications.

Rollo [20] explores the issues of measuring the size of Web applications using several approaches, discussed and compared later. All approaches examine size from the perspective of functionality. Rollo discusses difficulties and contradictions between these techniques and concludes, using empirical data, that the sizing method COSMIC-FFP[1] [2] is the most flexible approach for counting the functional size of Web sites and can be applied to any Web site. This work does not present a specific prediction model, instead explores the measurement of functional size. Although the empirical data was obtained measuring a Web site, this approach can also be used with hypermedia applications developed using other environments, e.g. Microcosm [3], Storyspace [4], VIKI [14] etc.

Mendes et al. [16] describe a case study involving the development of Web sites structured according to the Cognitive Flexibility [24] principles in which simple size metrics were collected. Several prediction models are generated (linear regression, stepwise regression and case-based reasoning) and compared. Results show that the best predictions were obtained using case-based reasoning, confirming similar results using non-hypermedia applications [21,23]. Their metrics can be applied to any hypermedia application, their method being an initial step towards the proposal of hypermedia development effort prediction models. However, some of these metrics are subjective, which may have influenced the validity of their results.

Fewster and Mendes [6] propose a prediction model for authoring and designing Web applications and for project risk analysis using a General Linear Model. They suggest two types of metrics: those applied to static Web applications and those applied to dynamic Web applications. The former metrics can also be used to measure other types of hypermedia applications. Their results were also an initial step towards the proposal of effort prediction models for hypermedia development.

CASE-BASED REASONING

Case-based Reasoning involves [21]:

- Characterising the project for which an estimate is required, i.e., identifying project attributes (application size, link complexity etc) which can influence effort.
- Use of this characterisation as a basis for finding similar (analogous) completed projects, for which effort is known.
- Use these effort values, possibly with adjustment, to generate a predicted value of effort.

Finding the right analogies and assessing the degree of similarity are the difficulties of this method.

When using CBR there are a number of issues to consider [23]:

- How to best characterise a project? The range of attributes must be limited to the information available at the point that the estimation is required. Attributes should portray the project as precisely as possible and it is essential to choose at least one attribute that characterises project in terms of size.
- How to detect similarities (analogies) between projects and with what level of confidence?
- How to identify the number of analogies to search for?
- How an effort estimate for a new project can be derived from analogous finished projects? Options are to use the mean effort of all analogous finished projects; to calculate a weighted mean of efforts where more influence is placed on the most similar finished projects (closer analogies).

Once there is a dataset containing historical data of finished projects, effort for new projects can then be estimated.

The analogies used in our work were calculated according to the following steps:

Step 1) Find the best combination of attributes to give the best estimate.

Step 2) Use the results obtained in step 1 to find the most similar finished project(s) to the new project, for which effort needs to be estimated. During this step it is also necessary to decide the number of projects (analogies) to be used in the estimation,

[1] COSMIC-FFP = COmmon Software Measurement International Consortium-Full Function Points

which can vary from 1 to total-number-of-completed-projects-available[2].

Step 3) Calculate the estimated effort as the mean of the known effort values for those projects selected in step 2[3].

A detailed description of CBR and a survey of several effort prediction models is given in [10,21].

THE ANGEL-PLUS SOFTWARE TOOL

It is feasible to find analogies, using the method described in Section 3, without the help of a computer program, however it can be both time consuming and error prone as the number of projects or attributes increase. Consequently, we have used a case-based reasoning effort estimation tool which automates the process and provides an environment where data can be stored, analogies found and estimates generated. The tool was developed at Bournemouth University under the name of ANaloGy softwarE tooL Plus (ANGEL-Plus). An important feature of ANGEL-Plus is the ability to determine the optimum combination of attributes for finding analogies.

Figure 1 (see appendix A) shows part of a template for recording project data. Templates are used to describe the data used in the estimation process and can be configured to match the specific data collection environment of an organisation. As ANGEL-Plus does not impose any particular set of attributes the estimation process can take optimal advantage of the data available. The user chooses the number of attributes and their names, the only exceptions are the project number, status and actual effort which are compulsory. ANGEL-Plus uses the project number to uniquely identify each project. Status shows whether or not a project is "on" or "off", representing respectively a completed project which can be used as a source of analogy or a new/ongoing project. The actual effort is mandatory for all completed projects as it supplies the foundation for prediction. Figure 2 (Appendix A) shows the data collected for the dataset used in our work.

Once the template has been created and data entered, the next step is to find the optimum combination of attributes used to find analogies. The justification behind the selection of attributes is that not necessarily all attributes collected will be useful to finding good analogies. The user can select all or a subset of attributes described in the template.

Once chosen, the attributes are used in the estimation algorithm, resulting in an estimated effort (see Figure 3 in Appendix A).

Typically where there are only a few projects or where the projects differ broadly in nature the results will not be as accurate as when the dataset represents similar projects.

Figure 3 (Appendix A) shows the estimated effort for a selected project, estimated based on three completed projects (3 analogies). The estimated effort is of 34 person-hours. Figure 3 also shows the best combination of attributes to generate the best estimate.

THE EXPERIMENT

The experiment [6] consisted of the design and authoring of hypermedia applications aimed at teaching a chosen topic, structured according to the Cognitive Flexibility Theory (CFT) principles [24], using a minimum of 50 documents. The delivery system (environment) chosen was the World Wide Web, as this was the one most students were familiar with.

Forty three Computer Science students from the University of Auckland participated in the experiment. They were attending the Hypermedia and Multimedia Systems course, offered during the first semester of 2000. Two questionnaires were used to collect the data. The first[4] asked subjects to rate their Web authoring experience using five scales, from no experience (zero) to very good experience (four). The second questionnaire[5] was used to measure characteristics of the Web applications developed (suggested metrics) and the effort involved in designing and authoring those applications. On both questionnaires, we describe in depth each scale type, to avoid any misunderstanding. Members of the research group checked both questionnaires for ambiguous questions, unusual tasks, number of questions and definitions in the Appendix.

Early analysis showed that six questionnaires were missing fundamental information. They were not used in the data analysis, leaving thirty seven datapoints.

To reduce learning effects, subjects were given a coursework prior to designing and authoring the Web applications, which consisted of:

- Creating a simple personal homepage.
- Designing a multi-page Web application.
- Creating a Web site for the Matakohe Kauri Museum[6], improving on their existing site.
- Loading the Web pages onto a Web server.

[2] Schofield [9] suggests that one project, followed by two projects, are generally the most effective options. However, it is possible that dissimilar datasets may reveal different characteristics.

[3] Although intuitively it would seem adequate to weight the mean in order to reflect the influence of the projects chosen in step 2, Kadoda et al. [10] show that using weighted mean does not contribute to the accuracy of a prediction.

[4] http://www.cs.auckland.ac.nz/~emilia/Assignments/exp-questionnaire.html.

[5] http://www.cs.auckland.ac.nz/~emilia/Assignments/questionnaire.html.

[6] http://www.hmu.auckland.ac.nz:8001/gilchrist/matakohe/.

Finally, all subjects received training on the Cognitive Flexibility Theory authoring principles for approximately 150 minutes.

Metrics collected

The metrics collected during the experiment (see Table 1) were organised into 3 categories:

- Size metrics - used to measure the size of hypermedia applications.
- Process metric - used to measure effort (person-hours).
- Confounding factors - used to measure possible factors that could influence the validity of the results.

Size Metrics	Description
Node Count	Number of html files.
Media Count	Number of not reused media files.
Reused Media Count	Number of reused[7] media files.
Total Node Allocation	Total space[8] allocated for all the html files.
Total Media Allocation	Total space[8] allocated for all the media files not reused.
Total Reused Media Allocation	Total space[8] allocated for all the reused media files.
Connectivity	Total number of internal links[9]. We do not include dynamically generated links.
Connectivity Density	*Connectivity* divided by *Node Count*.
Total Node Complexity	Average number of different types of media per node.
Cyclomatic Complexity	(*Connectivity* - *Node Count*) + 2.
Process Metric	**Description**
Effort	Estimated amount of person-hours to develop an application
Confounding Factors	**Description**
Structure	Main structure (backbone) of the application[10].
Experience	Authoring/design experience of a subject[11].
Type	Measures the type of tool used to author/design the Web pages[12]:

Table 1 - Size Metrics

[7] Reuse can be either black-box reuse or white-box reuse.
[8] Space was measured in Mbytes.
[9] Subjects did not use external links to other Web sites. All links pointed to pages within the application.
[10] Sequence, hierarchy or network.
[11] Measured using a scale from 0 (no experience) to 4 (a great deal of experience).
[12] WYSIWYG (What You See Is What You Get), close to WYSIWYG or text-based.

Validity of the Experiment

Here we present our comments on the validity of the experiment:

- The metrics collected, apart from effort, experience and structure, were all objective and precisely quantifiable. The scales used to measure *experience* and *structure* were described in detail in both questionnaires.
- Subjects' authoring and design experiences were mostly scaled little or average, with a low difference between skill levels. Consequently the original dataset was left intact.
- To reduce maturation effects, i.e. learning effects caused by subjects learning as an experiment proceeds, subjects had to develop each a Web application as part of their first assignment. They also received training in the CFT principles.
- All applications used a hierarchical structure.
- Notepad (or similar text editor) and FirstPage were the two tools most frequently used. Notepad is a simple text editor while FirstPage is freeware offering button-embedded HTML tags. Although they differ with respect to the functionality offered, data analysis revealed that the corresponding effort was similar, suggesting that confounding effects from the tools were controlled.
- As the subjects who participated in the experiment were either Computer Science Honours or MSc students, it is likely that they present skill sets similar to Web professionals at the start of their careers. The use of students as subjects, while sometimes considered unrealistic, is justified for two reasons: firstly, empirical evidence by Bohem-Davis and Ross [1] indicates that students are equal to professionals in many quantifiable measures, including their approach to developing software; secondly, for pragmatic considerations, having students as subjects was the only viable option for this experiment.

EMPIRICAL RESULTS USING CBR

In order to evaluate how precise the estimations were using CBR, we followed seven steps:

Step 1 One datapoint was chosen and removed from the dataset. This datapoint was considered as a new project for the sake of this evaluation.

Step 2 The remaining 36 datapoints were kept in the dataset and used in the estimation process.

Step 3 For the removed project, its actual effort data was discarded, and only the known size metrics kept. This was done in an attempt to simulate a new project for use with ANGEL-Plus to calculate estimated effort.

Step 4 The optimum combination of attributes was generated by using the "Best attribute set" option.

Step 5 Effort was estimated using 1 to 3 analogies, looking for the estimated effort closest to the removed project's discarded actual effort.

Step 6 The datapoint which had been removed on Step 1 was added back to the dataset.

Step 7 Steps 2 through 6 were repeated until all 37 projects had been removed and had had their effort estimated.

The results obtained (see Table 2) are as follows:

- Except for one datapoint, the best combination of attributes for the whole dataset was *Media Count*, *Total Node Allocation* and *Total Reused Media Allocation*, suggesting that the number of media files used by an application and the space allocated for nodes and reused media may be good indicators of effort. Further analysis of our dataset showed that there was a slight negative relationship between effort and both *Total Reused Media Allocation* and *Reused Media Count,* suggesting that the higher the amount of reuse the smaller the effort. The relationship between reuse and effort/productivity has received considerable attention in the literature [5,7,13,17,22] as it seems that level of reuse is inversely related to effort.
- The average percentage of error between actual effort and estimated effort was of 9.21%, indicating a very good prediction power [21].
- One analogy, followed by three analogies, presented the most accurate estimates.

Project	Actual Effort	EE1	EE2	EE3	PE Act&EE	Best combination of attributes
01	24.00	28	18	25*	4.17	Media Count + Total Node Allocation + Total Reused Media Allocation
02	32.00	40	36.5	34.3*	7.19	Media Count + Total Node Allocation + Total Reused Media Allocation
03	40.00	52	42.5	36*	6.25	Media Count + Total Node Allocation + Total Reused Media Allocation
04	46.00	35	43.5	43.3*	5.43	Media Count + Total Node Allocation + Total Reused Media Allocation
05	42.00	32	42	35.6*	0.00	Media Count + Total Node Allocation + Total Reused Media Allocation
06	62.00	45	45.5	57*	8.06	Media Count + Total Node Allocation + Total Reused Media Allocation
07	32.00	30*	34*	36.6	6.25	Media Count + Total Node Allocation + Total Reused Media Allocation
08	27.00	42	36*	40.3	33.33	Media Count + Total Node Allocation + Total Reused Media Allocation
09	39.00	39*	23	28.6	0.00	Media Count + Total Media Allocation
10	42.00	34*	24.5	32.6	19.05	Media Count + Total Node Allocation + Total Reused Media Allocation
11	49.00	42*	25	29.6	14.29	Media Count + Total Node Allocation + Total Reused Media Allocation
12	42.00	30	41*	30.6	2.38	Media Count + Total Node Allocation + Total Reused Media Allocation
13	52.00	48	40	45.3*	7.69	Media Count + Total Node Allocation + Total Reused Media Allocation
14	46.00	45	53.5	45.6*	0.87	Media Count + Total Node Allocation + Total Reused Media Allocation
15	50.00	52*	52*	37.3	4.00	Media Count + Total Node Allocation + Total Reused Media Allocation
16	26.00	28*	31.5	33.6	7.69	Media Count + Total Node Allocation + Total Reused Media Allocation
17	28.00	32*	41	41.6	14.29	Media Count + Total Node Allocation + Total Reused Media Allocation
18	20.00	39	23.5*	29	17.50	Media Count + Total Node Allocation + Total Reused Media Allocation
19	30.00	34	25	33*	10.00	Media Count + Total Node Allocation + Total Reused Media Allocation
20	29.00	27	26	27.3*	5.86	Media Count + Total Node Allocation + Total Reused Media Allocation
21	52.00	46*	16.5	25.3	11.54	Media Count + Total Node Allocation + Total Reused Media Allocation
22	40.00	30	40*	40.6	0.00	Media Count + Total Node Allocation + Total Reused Media Allocation
23	35.00	46	36*	38.3	2.86	Media Count + Total Node Allocation + Total Reused Media Allocation
24	45.00	46*	48	52.6	2.22	Media Count + Total Node Allocation + Total Reused Media Allocation
25	35.00	40*	41.5	46.3	14.29	Media Count + Total Node Allocation + Total Reused Media Allocation
26	48.00	52	54	50*	4.17	Media Count + Total Node Allocation + Total Reused Media Allocation
27	33.00	27	21	30.6*	7.27	Media Count + Total Node Allocation + Total Reused Media Allocation
28	52.00	50*	27	20.6	3.85	Media Count + Total Node Allocation + Total Reused Media Allocation
29	30.00	42	46	34*	13.33	Media Count + Total Node Allocation + Total Reused Media Allocation
30	28.00	24*	24*	18.3	14.29	Media Count + Total Node Allocation + Total Reused Media Allocation
31	34.00	42	15.5	26.6*	21.76	Media Count + Total Node Allocation + Total Reused Media Allocation
32	40.00	35	37*	34	7.50	Media Count + Total Node Allocation + Total Reused Media Allocation
33	30.00	48	39*	43.3	30.00	Media Count + Total Node Allocation + Total Reused Media Allocation
34	43.00	52	37.5*	32.3	12.79	Media Count + Total Node Allocation + Total Reused Media Allocation

35	39.00	39*	7.5	16.3	0.00	Media Count + Total Node Allocation + Total Reused Media Allocation
36	28.00	26*	45.5	47.6	7.14	Media Count + Total Node Allocation + Total Reused Media Allocation
37	40.00	32	31	34.6*	13.50	Media Count + Total Node Allocation + Total Reused Media Allocation
General PE Act&EE :					9.21	
EE1 = Estimated effort using 1 analogy **EE2** = Estimated effort using 2 analogies **EE3** = Estimated effort using 3 analogies **PE Act&EE** = Percentage of error between actual effort and estimated effort						

Table 2 – Results obtained using CBR

How can the results and prediction methodology presented in this paper be of benefit to hypermedia/Web developers?
In whatever type of hypermedia application (educational, fiction etc) there is at least one phase of their development which is common to all - authoring. Furthermore, most hypermedia systems reported in the literature provide environments to author static hypermedia applications [3,4,14]. Consequently, the metrics suggested in this paper may be of interest and useful to a wider range of hypermedia authors other than Web authors. In addition, the prediction methodology we have used can be applied to any organisation and to any type of hypermedia application and development environment.

In summary, there is an increased need for effective management of time and resources in organisations undertaking large numbers of projects and measurement principles can help understand, control and improve processes and products.

CONCLUSIONS AND FUTURE WORK

This paper presented the application of Case-based reasoning to estimating Web authoring and design effort. The dataset used was obtained through an experiment in which we measured Application, Process and Resource metrics.

The analogy was performed using an estimation tool - ANGEL-Plus. The average percentage of error between actual effort and estimated effort was of 9.21%, which represents very good predictions. This result suggests that Case-based reasoning is a candidate technique to effort estimation and that with the aid of an automated environment it is a practical technique to apply to hypermedia development effort prediction.

As part of our future work we plan to apply effort prediction to the whole development cycle of hypermedia applications and to compare the performance of case-based reasoning against human estimation and regression techniques, in a similar way to [8,16,18].

To conclude, there is an urgent need for adequate hypermedia development effort prediction at an early stage in the development. As the use of hypermedia, and in particular the Web, as a delivery environment increases, effort estimation can contribute significantly to the reduction of costs and time involved in developing hypermedia applications.

REFERENCES

[1] Boehm-Davis, D. A., and Ross, L. S., Program design methodologies and the software development process, International Journal of Man-Machine Studies, Vol. 36, pp. 1-19, Academic Press Limited, 1992.

[2] COSMIC, COSMIC-FFP Measurement manual, version 2.0, http://www.cosmicon.com, 1999.

[3] Davis, H., Hall, W., Heath, I., Hill, G., and Wilkings, R. Towards an Integrated Information Environment With Open Hypermedia Systems. In: Proceedings of the ACM Conference on Hypertext, 181-190, Milan, Italy, 1992.

[4] Dirberger, A., and Bolter, J. D. On the Design of Hyper Spaces. Communications of the ACM, Special Issue on Hypermedia Design, August, 1995.

[5] Fenton, N. E., and Pfleeger, S. L. Software Metrics, A Rigorous & Practical Approach, 2nd edition, PWS Publishing Company and International Thomson Computer Press, 1997.

[6] Fewster, R. and Mendes, E. Measurement, Prediction and Risk Analysis for Web Applications, Proc. IEEE Metrics Symposium, April, London, UK, 2000.

[7] Garzotto, F., Paolini, P., Baresi, L., Supporting Reusable Web-design with HMD-Edit, Proceedings of the Thirty-fourth Annual Hawaii International Conference on System Sciences, 2001, pp. 1-10.

[8] Gray, R. and MacDonell, S. G. A Comparison of Techniques for Developing Predictive Models of Software Metrics, Information and Software Technology, Vol. 39, 1997, pp. 425-437.

[9] Hatzimanikatis, E., Tsalidis, C. T. and Christodoulakis, D. Measuring the Readability and Maintainability of Hyperdocuments, J. of Software Maintenance, Research and Practice, Vol. 7, 1995, pp. 77-90.

[10] Hughes, R.T. An Empirical Investigation into the Estimation of Software Development Effort, PhD Thesis, University of Brighton, UK, 1997.

[11] Kadoda, Gada, Cartwright, Michelle, Chen, Linguang, and Shepperd, Marting, Experiences Using Case-Based Reasoning to Predict Software Effort, Proceedings of the EASE200 - Conference on Empirical Assessment In Software Engineering, April 2000, pp. 1-17.

[12] Kitchenham, B., Pickard, L., and Pfleeger, S. L. Case Studies for Method and Tool Evaluation, IEEE Software, July, 52-62, 1995.

[13] Lowe, D., and Hall, W. Hypertext and the Web - An Engineering Approach, John Wiley & Sons Ltd., eds., 1998.

[14] Marshall, C., Shipman, F. M. III, and Coombs, J. H. VIKI: Spatial Hypertext Supporting Emergent Structure. In: Proceedings of ECHT'94, Edinburgh, September, 1994.

[15] McDonell, S. G., and Fletcher, T. Metric Selection for Effort Assessment in Multimedia Systems Development, Proc. Metrics'98, 1998.

[16] Mendes, E., Counsell, S., and Mosley, N., Measurement and Effort Prediction of Web Applications, Proc. Second ICSE Workshop on Web Engineering, 4 and 5 June 2000; Limerick, Ireland, 2000.

[17] Morisio, M., Stamelos, I., Spahos, V. and Romano, D., Measuring Functionality and Productivity in Web-based applications: a Case Study, Proceedings of the Sixth International Software Metrics Symposium, 1999, pp. 111-118.

[18] Myrtveit, and Stensrud, E., A Controlled Experiment to Assess the Benefits of Estimating with Analogy and Regression Models, IEEE Transactions on Software Engineering, Vol. 25, No. 4, Jul./Aug. 1999, pp. 510-525.

[19] Pfleeger, S. L., Jeffery, R., Curtis, B., and Kitchenham, B. Status Report on Software Measurement, IEEE Software, March/April, 1997.

[20] Rollo, Tony, Sizing E-Commerce, Proceedings of the ACOSM 2000 - Australian Conference on Software Measurement, Sydney, 2000.

[21] Schofield, C. An empirical investigation into software estimation by analogy, PhD Thesis, Dept. of Computing, Bournemouth Univ., UK, 1998.

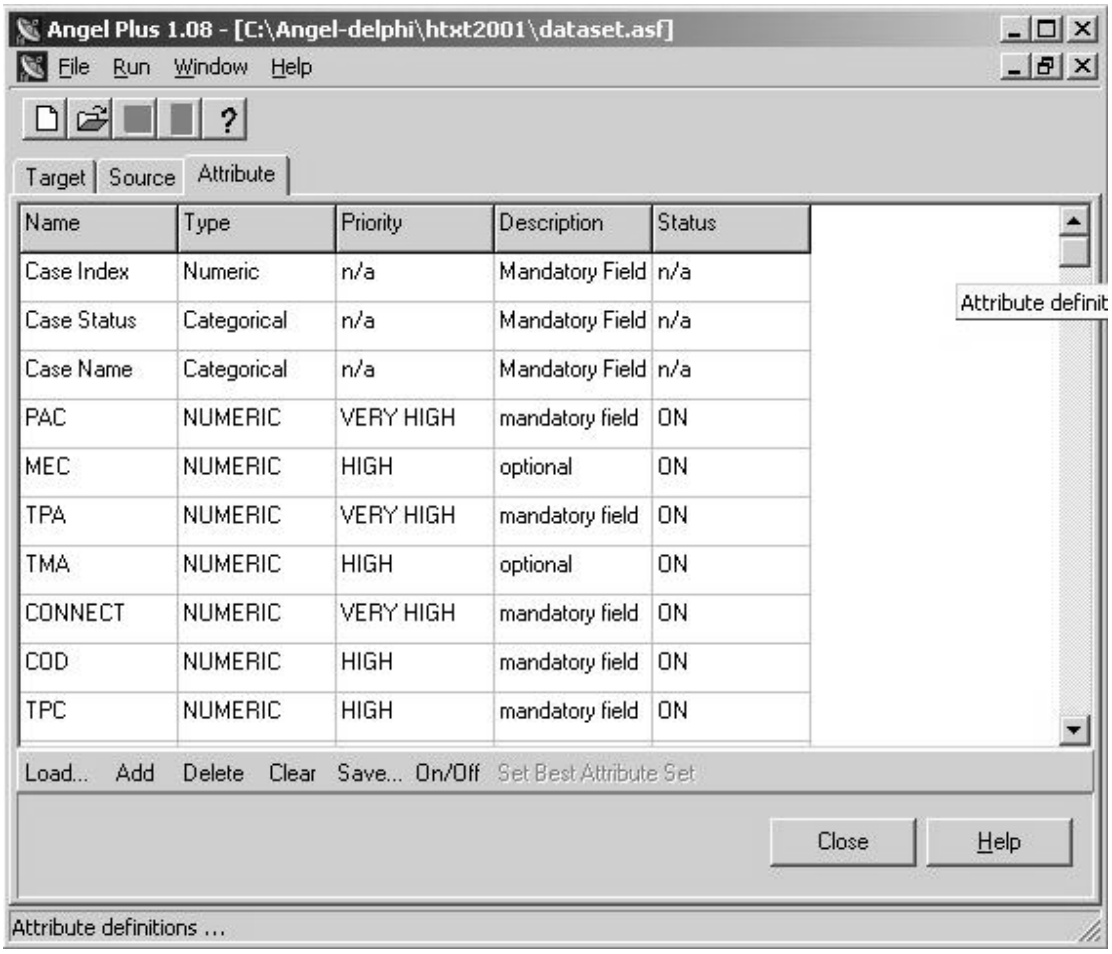
[22] Schwabe, D., Esmeraldo, L., Rossi, G., and Lyardet, F. Engineering Web Applications for Reuse, IEEE Multimedia, January-March, 2001, pp. 20-31.

[23] Shepperd, M.J., Schofield, C., and Kitchenham, B., Effort Estimation Using Analogy. Proc. ICSE-18, IEEE Computer Society Press, Berlin, 1996.

[24] Spiro, R. J., Feltovich, P. J., Jacobson, M. J., and Coulson, R. L. Cognitive Flexibility, Constructivism, and Hypertext: Random Access Instruction for Advanced Knowledge Acquisition in Ill-Structured Domains, In: L. Steffe & J. Gale, eds., *Constructivism*, Hillsdale, N.J.:Erlbaum, 1995.

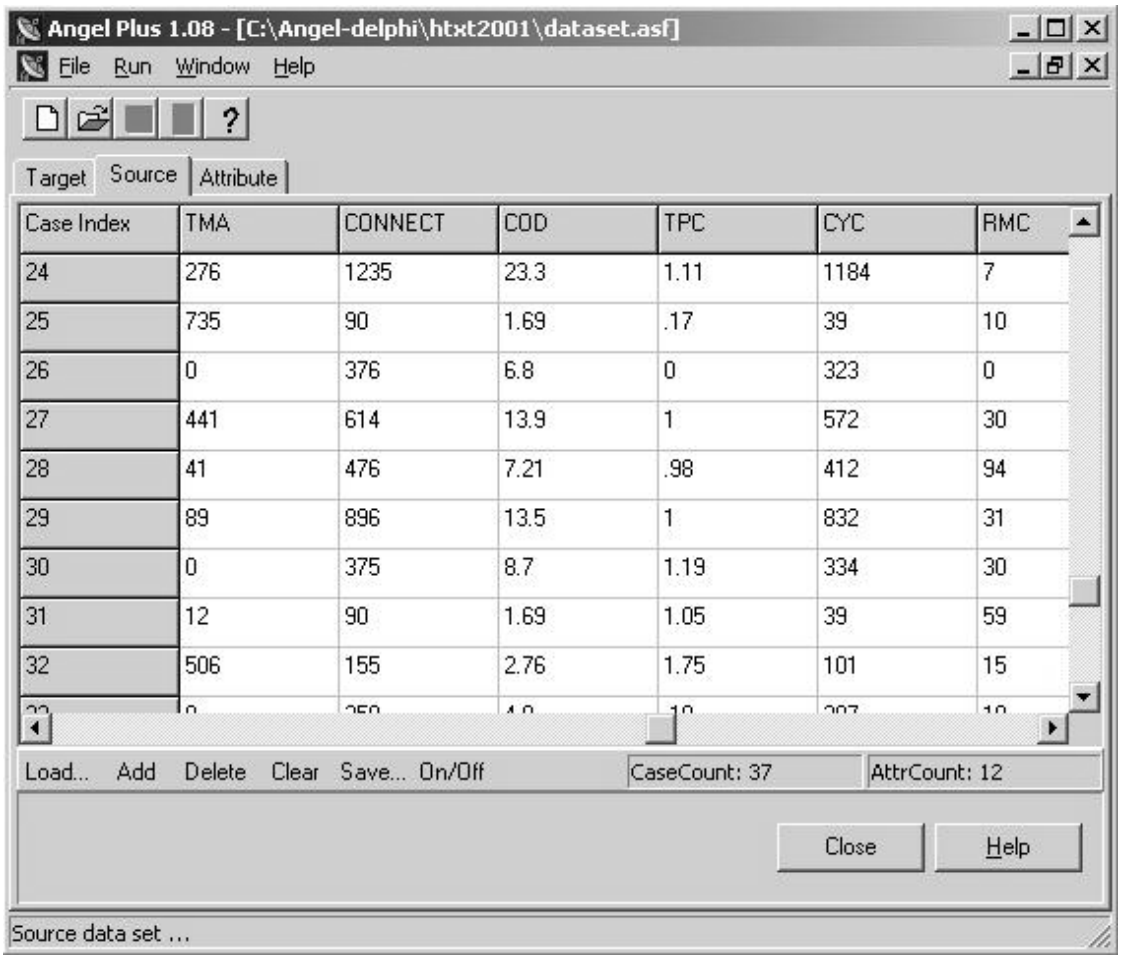
[25] Warren, P., Boldyreff, C., and Munro, M. The Evolution of Websites, Proc. Seventh International Workshop on Program Comprehension, IEEE Computer Society Press, Los Alamitos, Calif., 1999, pp. 178-185.

APPENDIX



Name	Type	Priority	Description	Status
Case Index	Numeric	n/a	Mandatory Field	n/a
Case Status	Categorical	n/a	Mandatory Field	n/a
Case Name	Categorical	n/a	Mandatory Field	n/a
PAC	NUMERIC	VERY HIGH	mandatory field	ON
MEC	NUMERIC	HIGH	optional	ON
TPA	NUMERIC	VERY HIGH	mandatory field	ON
TMA	NUMERIC	HIGH	optional	ON
CONNECT	NUMERIC	VERY HIGH	mandatory field	ON
COD	NUMERIC	HIGH	mandatory field	ON
TPC	NUMERIC	HIGH	mandatory field	ON

Figure 1 – Data Entry Templates in ANGEL-Plus



Case Index	TMA	CONNECT	COD	TPC	CYC	RMC
24	276	1235	23.3	1.11	1184	7
25	735	90	1.69	.17	39	10
26	0	376	6.8	0	323	0
27	441	614	13.9	1	572	30
28	41	476	7.21	.98	412	94
29	89	896	13.5	1	832	31
30	0	375	8.7	1.19	334	30
31	12	90	1.69	1.05	39	59
32	506	155	2.76	1.75	101	15

Figure 2 – A Dataset in ANGEL-Plus

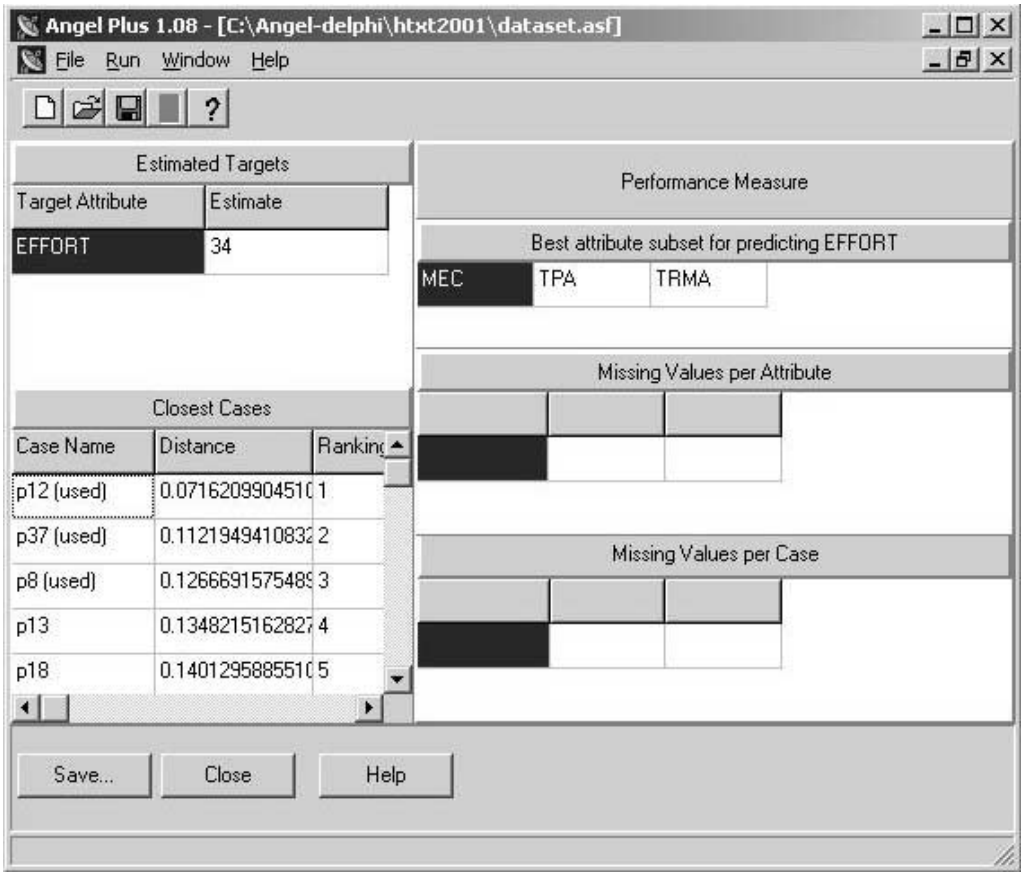


Figure 3 – Predicting Effort using ANGEL-Plus